# Diffraction

**antocciphysics**

**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Explain spreading around edges and through openings.

**SUCCESS CRITERIA - I can...**

1. Compare opening size with wavelength.
2. Distinguish diffraction from refraction.
3. Use a wave picture to justify spreading.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

**LANGUAGE GOAL** Use “diffraction” and “wavelength” to explain your model or result. Sentence starters are optional.

**diffraction**: wave spreading around an edge or through an opening

**wavelength**: distance between corresponding points on repeated waves

**boundary**: the place where a medium or constraint changes

## 01 Bring back an earlier idea

A wave travels 9 m in 0.3 s at frequency 50 Hz.
**Find** speed and wavelength.

**Word help:** frequency = number of cycles each second

**Sentence starter:** I found ... [unit]. This value means ... in this situation.

**Marzano check - circle:** ① ② ③

## 02 Explain a worked example

**Reference:** λ = v/f; compare opening width with λ. Spreading is pronounced when the opening width is comparable to wavelength. Label the opening and wavefronts.

**Worked reasoning:** A wave with wavelength 1 m meets openings 1 m and 10 m wide. Spreading relative to opening size is more noticeable for the 1 m opening.

**Your turn: Draw** incoming parallel wavefronts and the emerging patterns for each opening.

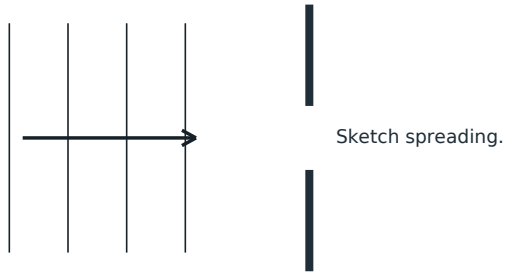
**Sentence starter:** The ... in my model represents ... . Its label shows ... .

**Marzano check - circle:** ① ② ③

## 03 Represent and interpret

**Sketch** wavefronts after a narrow opening about one wavelength wide.
**Compare** with a much wider opening.

**Sentence starter:** Both ... . They differ in ... because ... .




**Marzano check - circle:** ① ② ③

## 04 Solve and justify

A sound wave has frequency 340 Hz and speed 340 m/s. Findλ and compare a .8 m doorway with a .0000005 m visible-light wavelength. Why does heard-around-corner behavior differ?

**Sentence starter:** Both ... . They differ in ... because ... .

**Marzano check - circle:** ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

## 05 Analyze the reasoning

A student calls all wave bending “refraction.” Explain how diffraction can happen without crossing into a different-speed medium.

**Sentence starter:** Both … . They differ in … because … .

**Marzano check - circle:** ① ② ③

## 06 Choose a method

For a fixed ideal-gas particle mass, average translational kinetic energy becomes 16 times as large.
**Compare** rms speed.

**Word help:** particle = a small constituent, such as an atom or molecule

**Sentence starter:** Both … . They differ in … because … .

**Marzano check - circle:** ① ② ③

## 07 Exit check

At fixed wave speed, frequency doubles.
**Explain** howλand typical diffraction through a fixed opening change.

**Sentence starter:** Under the assumption …, the evidence … supports … because … .

**Marzano check - circle:** ① ② ③

## 08 STRETCH: Apply it another way

Sound speed is 340 m/s.
**Compare** wavelengths at 170 and 1700 Hz with a 1 m doorway.
**Explain** which should diffract more noticeably.

**Sentence starter:** Both … . They differ in … because … .

**Marzano check - circle:** ① ② ③